Hypercom®, Inc.
2851 West Kathleen Road
Phoenix, Arizona 85023
Ph: (602) 866-5399
Fax: (602) 866-5380

HYPERCOM

T7 / FIP11

ECR INTERFACE

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NOTICE

TABLE OF CONTENTS

1. DOCUMENT MODIFICATION HISTORY

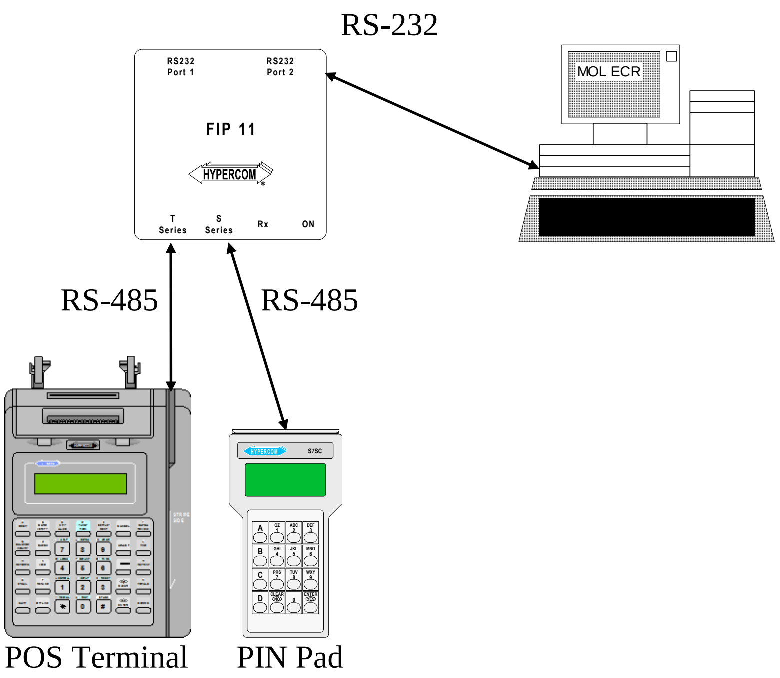
Date	Revision	Change
May 26, 1995	1.00	Initial Release
June 12, 1995	1.01	Modify message flow so initial message from ECR contains transaction totals
June 14	1.02	Provide more detail for physical interface section and correct the baud rate defined. Add more detail to the link and message leve descriptions.
June 16, 1995	1.03	Update ACK/NAK procesing description. Correct ECR timeout processing.
March 3, 2010	1.04	Masking of exp date and PAN was added
December 20, 2010	1.05	Settlement and VOID Message IDs were added
December 22, 2010	1.06	Message flow was corrected for STL transaction; DIS – Discount message was added;

2. OVERVIEW

This document describes the software interface between a Hypercom POS terminal and a MOL Electronic Cash Register (ECR).

3. PHYSICAL INTERFACE

The Physical Interface to the Hypercom terminal will be implemented by use of the PIN pad full-duplex RS-485 port. A FIP-11 will be used to provide connection for a Hypercom PIN pad and a RS-232 serial interface as illustrated below:



The connection between the FIP-11 and the terminal is a straight RJ-11 cable. The connection between the FIP-11 and the PIN pad is a reverse RJ-11 cable. The pin connections of the FIP-11 RS-232 port is defined as follows:

Signal Name	Pin Number	Direction
RD	2	⇐
TD	3	⇒
DTR	4	⇒
GND	5	
DSR	6	⇐
RTS	7	⇒
CTS	8	⇐

Power is supplied to the PIN pad and FIP-11 by the POS terminal.

Data transmission is Asynchronous at 9600 BPS, with 7 data bits, even parity, and 1 stop bits. 7 bit ASCII characters are supported with least significant bit transmitted first; parity is most significant bit.

4. LINK LEVEL PROTOCOL

4.1. Message block format

STX	MESSAGE BLOCK	ETX	LRC
1	variable, 128 maximum	1	1

4.2. Communications Characters Used

Character	Hex Value	Meaning
STX	02	Start of text
ETX	03	End of Text
EOT	04	End Of Transmission
ACK	06	positive ACKnowledgment
NAK	15	Negative AcKnowledgment

4.3. LRC - Longitudinal Redundancy Check

The LRC character provides an error detection facility.

The LRC is generated by Exclusive-OR'ing all data bytes <u>following</u> the STX, up to and <u>including</u> the ETX.

The LRC byte follows the ETX, and must have the same parity as the rest of the message.

4.4. ACK/NAK Responses

The FIP-11 sends all messages received from the terminal to both the PIN pad port and the ECR serial port. The PIN pad and the ECR must only ACK or NAK messages that are intended for the specific device. The external devices must inspect the message content to determine if the message type is for the PIN pad or the ECR and generate an ACK or NAK only when the message is acceptable to the device. Failure to decode the message by any device will result in no ACK or NAK and the terminal will retransmit the message automatically.

The terminal will receive messages from the external devices and will ACK or NAK as required. The PIN pad and ECR must ignore unexpected ACK or NAKs when not actually performing message transactions.

4.5. Positive Request/Response Transmission

MOL ECR/POS TERMINAL	⇔	MOL ECR/POS TERMINAL
STX Message ETX LRC	⇒	
	⇐	ACK

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- Minimum ACKnowledgment time = 0ms.
- Maximum ACKnowledgment time = 1000ms (Recommended 1500ms between last ACK and sequence “13”).
- If no acknowledgment is received before the time expires then it is treated as a NAK and the message is retransmitted.

4.6. Request/Response Transmission with Retries

MOL ECR/POS TERMINAL	⇔	MOL ECR/POS TERMINAL
STX Message ETX LRC	⇒	
	⇐	NAK
STX Message ETX LRC	⇒	
	⇐	NAK
STX Message ETX LRC	⇒	
	⇐	NAK

- Maximum number of retries = 3.

5. MESSAGE PROCESSING

5.1. Message Definitions

- All fields use ASCII character set.

5.1.1. (10) Initiate Transaction

The ECR will generate this message to the terminal. When the terminal receives this transaction it will initiate the transaction process and prompt the user to swipe the card. The transaction total will be included in this message and all other data required will be prompted on the terminal. The terminal will acknowledge the request prior to processing the transaction.

Request

NAME	LENGTH	VALUE	DESCRIPTION
STX	1	02h	Start of Text
Message ID	3	XXX	Message Type Identifier, MOL = Purchase, VOI = VOID, STL = Settlement, DIS = Discount
Message Type	2	10	Initiate Financial Transaction Request
Separator	1	.	Separator Character
Transaction Amount	12	X.....X	Transaction total amount, without decimal point (right justified, unused places are fitted by zeros)
ETX	1	03h	End of Text
LRC	1	xxh	Longitudinal Redundancy Character

- The transaction amount is 12 digits with an implied decimal as determined by local currency. For VOID transactions this field is used for transmitting of invoice number of transaction which should be voided;

Response

NAME	LENGTH	VALUE	DESCRIPTION
STX	1	02h	Start of Text
Message ID	3	XXX	Message Type Identifier, MOL = Purchase, VOI = VOID, STL = Settlement, DIS = Discount
Message Type	2	11	Initiate Financial Transaction Response
Separator	1	.	Separator Character
ETX	1	03h	End of Text
LRC	1	xxh	Longitudinal Redundancy Character

5.1.2. (12) Authorization Response

Request

The terminal will generate this message to the ECR. This message is generated after the financial transaction is completed. If the transaction is denied then all fields will be empty except the response code.

NAME	LENGTH	VALUE	DESCRIPTION
STX	1	02h	Start of Text
Message ID	3	XXX	Message Type Identifier, MOL = Purchase, VOI = VOID, DIS = Discount
Message Type	2	12	Authorization Response
Separator	1	.	Separator Character
Response Code	2	XX	Authorization response code, 0 = approval
PAN	1-18	X.....X	Account Number (including check digit)*
FS	1	1Ch	Field Separator
Card Expiry Date	4	XXXX	Card expiry date, YYMM format*
Invoice Number	6	XXXXXX	Transaction invoice number, right justified, zero fill
Approval Code	6	XXXXXX	Host approval code, right justified, zero fill
Date	6	XXXXXX	Transaction date, YYMMDD
Time	6	XXXXXX	Transaction time, HHMMSS
Issuer Name	1-8	X.....X	Card issuer name
FS	1	1Ch	Field Separator
ETX	1	03h	End of Text
LRC	1	xxh	Longitudinal Redundancy Character

- The authorization response code is a 2 digit value as defined by the ISO-8583 standards. This will be the value returned by the host processor authorizing the transaction.
- The PAN is a 1-18 digit account number assigned to the card used for the transaction.
- The issuer name is the value assigned to the issuer when the Term-Master configuration is defined. This issuer that is used will be determined by the POS terminal dependent on the range of the account number.
- Card Expiration Date is transmitted depend on "Exp. Date Mask" option from Acquirer Table. If option was selected POS transmits "****" otherwise Expiration Date (MMYY)
- The field can be sent in masked form like PAN on the receipt (see description of PAN Mask field in acquirer table)
- For STL command Authorization Result (12) is not used, because of Settlement transaction can take a long time.

Response

NAME	LENGTH	VALUE	DESCRIPTION
STX	1	02h	Start of Text
Message ID	3	XXX	Message Type Identifier
Message Type	2	13	Authorization acknowledgment
Separator	1	.	Separator Character
ETX	1	03h	End of Text
LRC	1	xxh	Longitudinal Redundancy Character

5.2. Message Transmission

5.2.1. Normal Message Sequence

MOL ECR	⇔	POS TERMINAL
Initiate Transaction (10)	⇒	
	⇐	ACK
	⇐	Initiate Response (11)
ACK	⇒	
	⇐	Authorization Result (12)
ACK	⇒	
Authorization Response (13)	⇒	
	⇐	ACK

5.2.2. Normal Message Sequence for STL command

MOL ECR	⇔	POS TERMINAL
Initiate Transaction (10)	⇒	
	⇐	Initiate Response (11)
ACK	⇒	
	⇐	ACK

5.2.3. Abnormal Message Sequence Processing

- If the Initiate Transaction (10) is not ACKed by the POS terminal or the Initiate Response is not received by the ECR within 30 seconds then the ECR will assume the terminal is off-line and the transaction completed as a stand-beside operation.
- If the POS terminal does not receive an ACK for the Initiate Response (11) within 30 seconds of transmitting the Initiate Response (11) then the terminal will abort the transaction.
- The ECR will start a configurable timer when the Initiate Response is received from the POS terminal. This timer should have a minimum value of 60 seconds and should provide enough time for the user to perform the terminal operations, execute the authorization call, and return the Authorization Result. If the POS terminal does not transmit the Authorization Result (12) within the time configured on the ECR, then the ECR will assume the terminal has gone off-line. Note that the terminal may have completed the transaction and printed a receipt

but was unable to return notification to the ECR. The ECR will provide a method of manually completing the sale in this event.

- If the POS terminal does not receive the ACK or Authorization Response (13) within 30 seconds of attempting to send the Authorization Result (12) then the terminal will return to the quiescent state waiting for the next transaction.

5.3. Message Parsing Errors

The FIP-11 sends all messages received from the terminal to both the PIN pad port and the ECR serial port. The ECR will therefore receive messages that are intended for the PIN pad and must not respond with an error messages when the message is not recognized. The ECR should only respond to the messages defined in this document to prevent conflict with PIN pad communications.